The Knee: Breaking llama.cpp Vulkan's 8-Thread Ceiling

How a single line of code was masking a 6.7x performance cliff—found, proven, and patched.



[9TH_THREAD_AGGREGATE]
19.7 ➔ **131.5 tok/s** (x6.7)

[10_CONCURRENT_LATENCY]
204s ➔ 38s (x5.4)

[HARDWARE_COST]
$0.00 (Zero silicon required)

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A Software Wall Wearing a Hardware Costume



The Symptom: Multi-user deployments died exactly at the 9th concurrent stream. Per-stream rates dropped to 1.3 tok/s (below human reading speed).

The False Attribution: Blamed on VRAM, memory bandwidth, and host limits. It survived a complete machine rebuild (new CPU, 4x RAM, new OS, new driver) completely untouched.

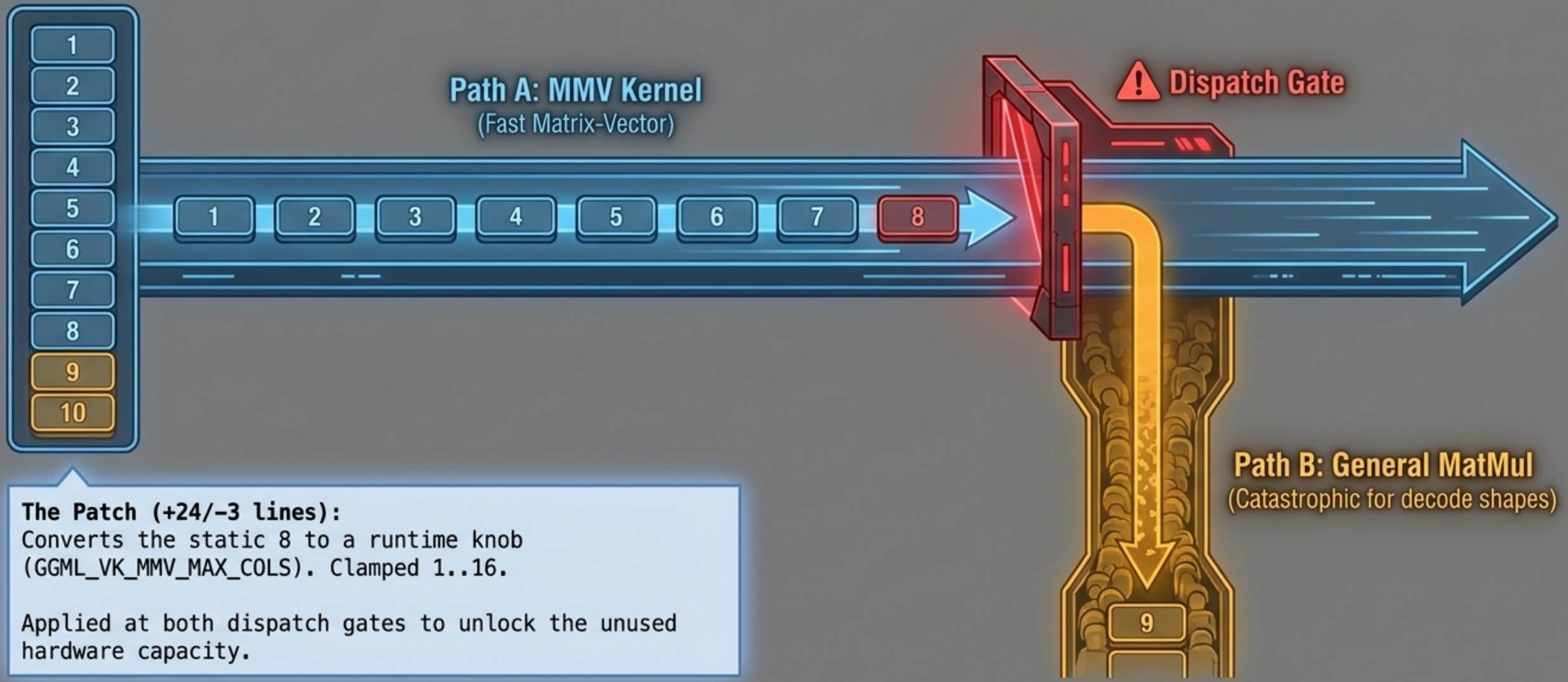
```
static constexpr uint32_t
 mul_mat_vec_max_cols = 8;
```

The Truth: ggml-vulkan.cpp line 389. A static compile-time constant blind to silicon.

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The Dispatch Gate Error

Decode batches of ≤ 8 sequences ride the highly optimized MMV path. The 9th sequence forces the entire batch onto a general matrix-multiply fallback, collapsing aggregate throughput by ~6x.



The Patch (+24/−3 lines):
Converts the static 8 to a runtime knob (GGML_VK_MMV_MAX_COLS). Clamped 1..16.

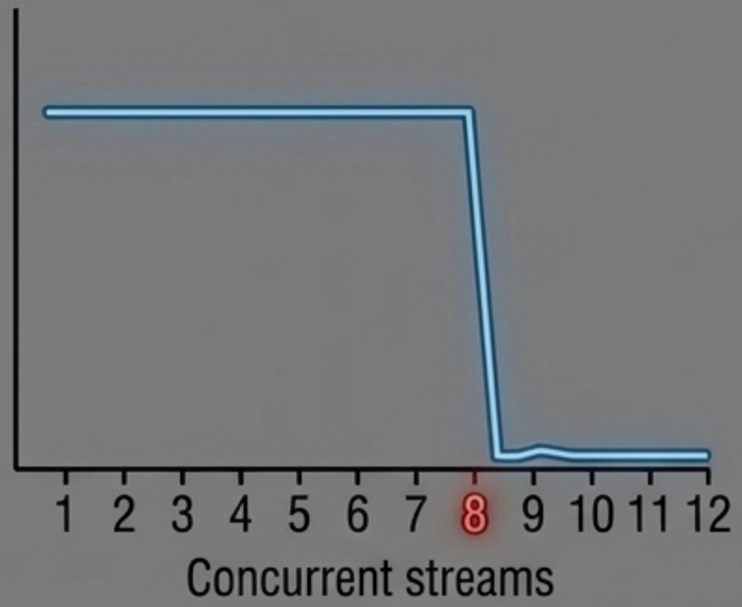
Applied at both dispatch gates to unlock the unused hardware capacity.

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The Host-Invariant Cliff

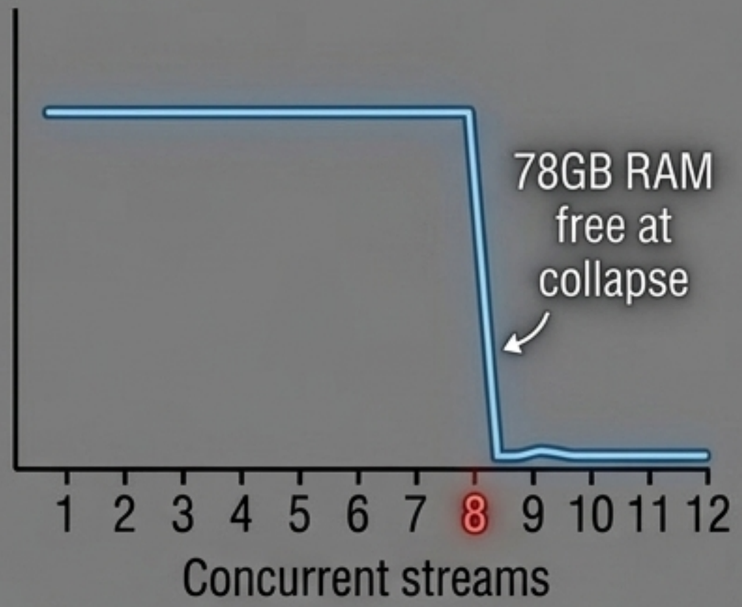
Rig 1 (May 2026)

Ryzen 9 5900X (AM4), 32GB DDR4, Dual Arc B70



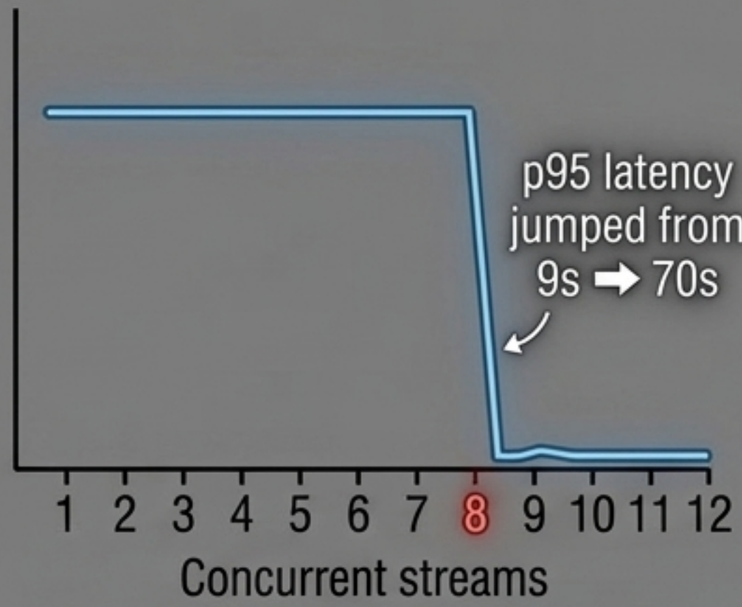
Rig 2 (Aug 2026)

Intel Core Ultra 285K, 128GB DDR5, Dual Arc B70



Rig 3 (The Discriminator)

Single Arc B70, dense 14B model, single tenant

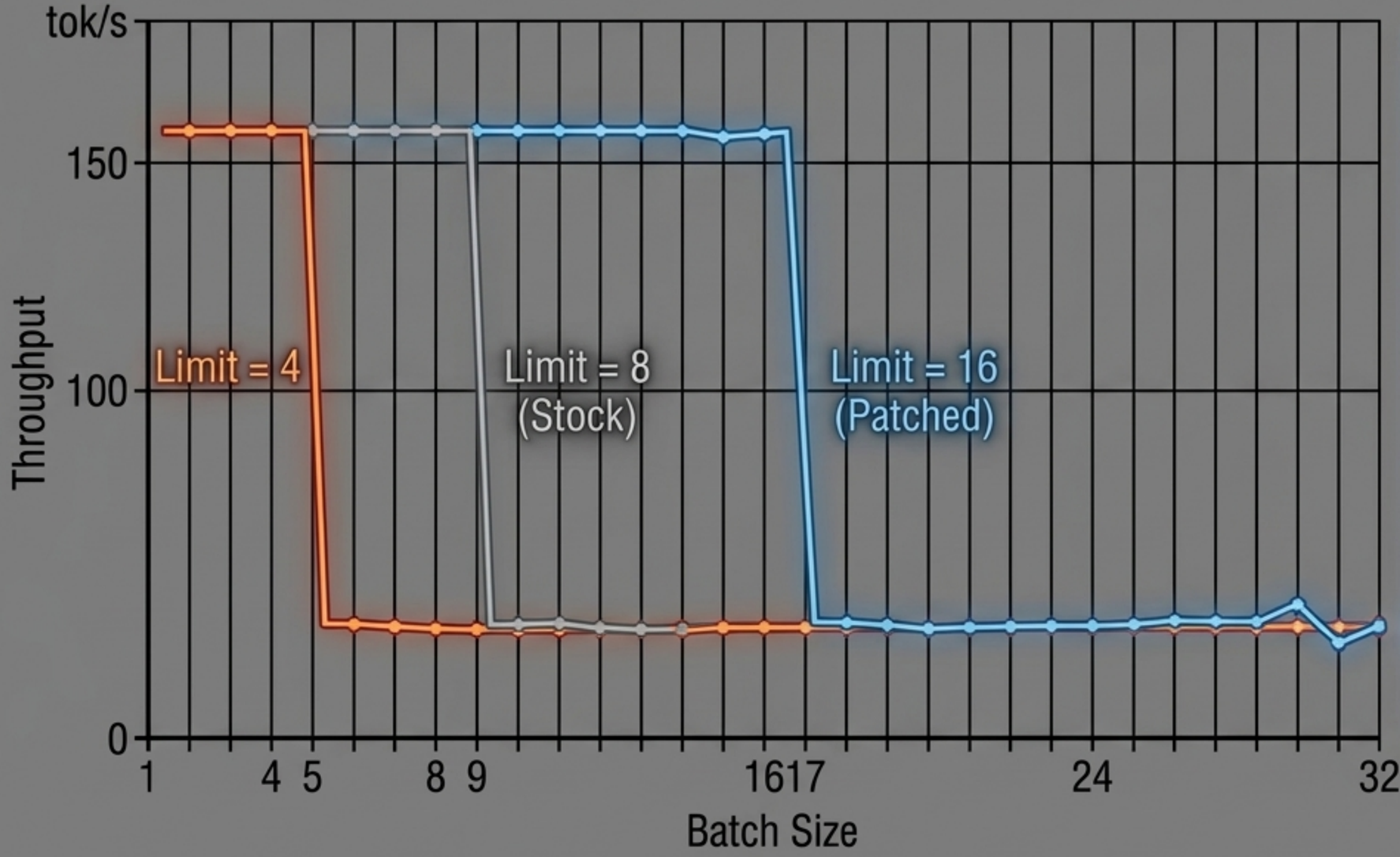


The bandwidth roofline should move with hardware and bytes/token.
This knee never moved. The limit lived exclusively in software.

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Causality Proven by Dose-Response

The Experiment: One binary, one variable changed (GGML_VK_MMV_MAX_COLS). Mistral-Small-3.2-24B, cool-start gated, zero VRAM spill.

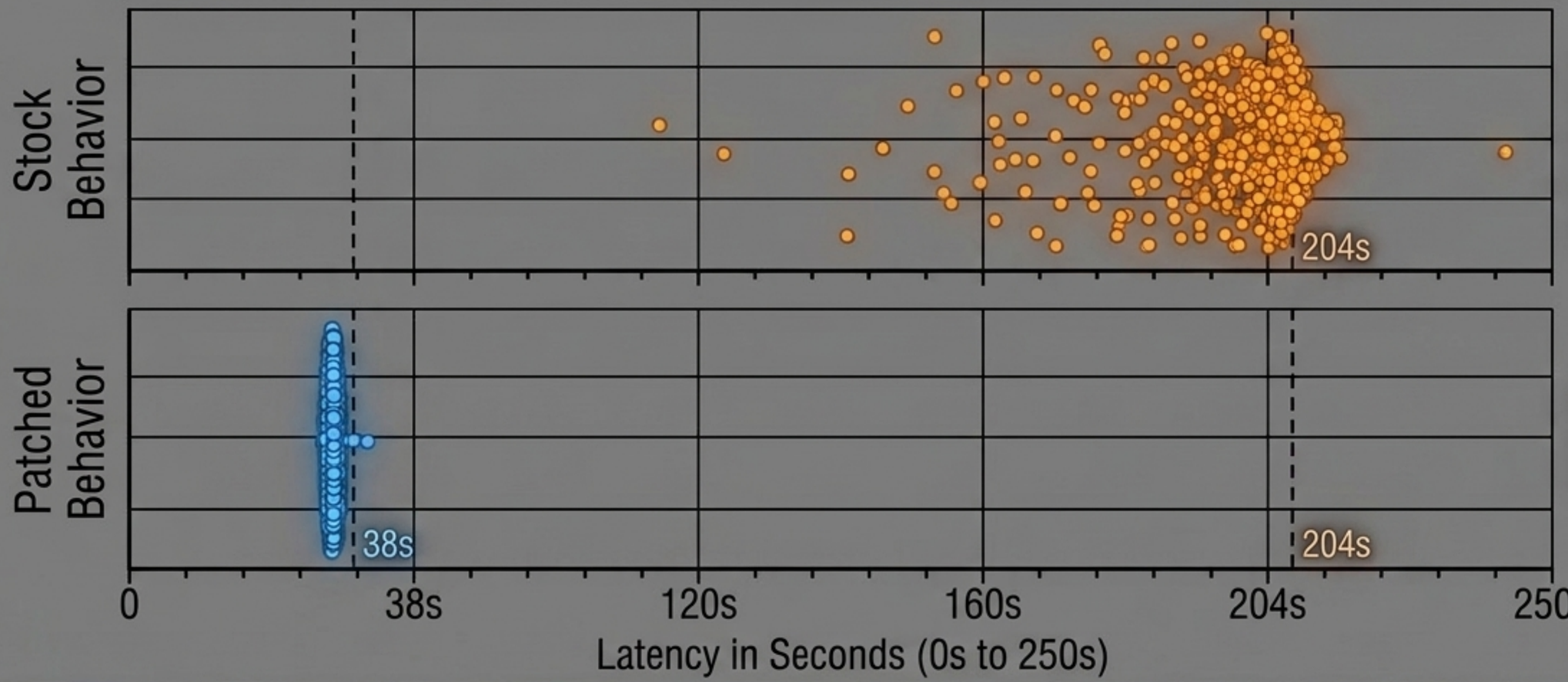


- We move the variable; the physical cliff follows perfectly (4 ➡ 6, 8 ➡ 9, 16 ➡ 24).
- Off-cliff values agree across all limits to within noise.
- The Inversion: The MMV fast-path at B=16 (158.5 tok/s) physically beats the heavy MatMul path at B=32 (139.5 tok/s).

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Threads-in-Flow: Experiential Impact

Test: 10 concurrent requests (Mistral-Small-24B, n_predict=200).



The Bottom Line: Moving 8 ➔ 16 roughly doubles the parallel-agent capacity of the exact same silicon at zero hardware cost.

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Impact Radius: Dense vs. MoE Architecture

Dense Architecture Models (Full 6.6x - 6.9x gain at B=9)	
Mistral-Small-24B	Gain: x6.62
Qwen2.5-32B	Gain: x6.88
Gemma-3-27B	Gain: x6.84
Phi-4 14B	Gain: x6.79

MoE Architecture Models (Fractional 1.1x - 1.8x gain)	
gpt-oss-20b	Gain: x1.82
Qwen3-30B	Gain: x1.13

Architectural Insight: MoEs are hit on dense attention projections only. Their expert FFNs sit behind a completely separate dispatch gate (MUL_MAT_ID), insulating them from the MMV cliff but capping their total gain.

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Mathematical Equivalence & Rigor

CPU Reference Test

[PASS]

`test-backend-ops.exe -o MUL_MAT -b Vulkan0`

Exercises new MMV bounds (cols 9-16) against CPU reference. 100% pass rate with tolerance checks.

Stock Equivalence

[PASS]

The new default (8) is a true no-op.

Matches b10549 prebuilt binaries to within ±3% margin. No behavioral regressions.

End-to-End Perplexity

[PASS]

Deterministic 472k-byte evaluation on Mistral-24B.

```
Stock:   PPL = 6.3139 +/- 0.28430
Patched: PPL = 6.3139 +/- 0.28430
```

Identical to every printed digit. Chunked evaluation ran 1.6x faster.

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PR #25356: Ready for Upstream

The Implementation is Safe: Deployed as an environment knob (GGML_VK_MMV_MAX_COLS). Conservative default of 8 means zero blast radius for untested setups.
Cross-Platform Alignment: Matches the exact shape of the CUDA fix merged in August 2026 (PR #26079, GGML_CUDA_MMVQ_MAX).
Upstream Issue: #25356 Data Submitted.

[x] Issue #25356 Data Submitted

[x] Zero TDR/WHEA events across burn-in

[x] Lab verified & AI disclosed per CONTRIBUTING.md

[x] PR branch cut: vulkan: add GGML_VK_MMV_MAX_COLS override

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